



Supplementary material for “A closed-form reference distribution for goodness-of-fit testing under penalised logistic regression in the proportional regime”

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Abstract

This document reports the material referred to in the paper: the behaviour of the correction under penalties other than the ridge.

Keywords calibration; goodness-of-fit; penalised regression

S1. Penalties other than the ridge

Nothing in Sections 2–4 of the paper uses the linearity of the ridge penalty. If a smooth penalty contributes a score term $a(\beta)$ with $K(\beta) = -\partial a/\partial \beta$, then $\tilde{\beta} = \hat{\beta} - F^{-1}a(\hat{\beta})$, $\hat{\mu} = -UM^{-1}a(\hat{\beta})$ and Theorem 1 of the paper go through with K evaluated at $\tilde{\beta}$ (Ebrahim, 2026, Remark 2). We implemented this for the generalised ridge penalty $\sum_j \theta_j \beta_j^2$ and for Firth's penalty (Firth, 1993), whose score contribution is the Jeffreys term $a_j(\beta) = \frac{1}{2} \text{tr}(F^{-1} \partial F / \partial \beta_j)$ and which is the standard response to separation in clinical prediction (Puhr et al., 2017); the Firth fit agrees with the `logistf` package (Heinze and Schemper, 2002) to 2×10^{-8} .

Table S1 makes three points. Under the generalised ridge the picture is the ridge picture: the uncorrected test rejects almost every correct model, the first-order reference is conservative by the deflation of Theorem 1 of the paper (ratio 0.91 at fixed p and 0.83 at $\kappa = 0.25$), and the variance at the debiased fit restores the level at fixed p (0.060 and 0.047) and leaves at $\kappa = 0.25$ the same two-to-three-point excess (0.080 and 0.070) that Section 4 of the paper treats. Under Firth's penalty at fixed p there is nothing to correct: the Jeffreys term is $O(n^{-1})$, the displacement is negligible, and the three references coincide at 0.067 on the decile basis and 0.053 on the EDGE basis. At $\kappa = 0.25$ Firth's penalty is not a shrinkage penalty. The fitted coefficient norm is 3.51 against a true value of 2.31, the proportional-regime inflation of Sur and Candès (2019) that an $O(n^{-1})$ bias reduction does not touch, and the one-step debiasing moves further away from the truth (4.75), so the corrected references reject 12 to 32 per cent of correct models and the uncorrected test 99 per cent. The framework of this paper is for estimators that shrink towards zero. For bias-reduced estimators in the proportional regime the reference would have to be built on the inflation rather than on the displacement, and we leave that open.

Table S1. Rejection rate of a correctly specified model at nominal level 0.05 under three smooth penalties, designs A ($\lambda = 100$) and B ($\kappa = 0.25$, $\lambda = 416$; the generalised ridge uses $\theta_j = \lambda(1 + j/p)$): the uncorrected statistic under the maximum likelihood covariance, the corrected statistic under the first-order reference Ω_{MLE} , and under the variance at the debiased fit Σ_d ; last column, the deflation ratio $E(SC.HL)/\text{tr}(\Omega_{MLE})$. 300 replications per cell.

Design	penalty	Decile basis			EDGE basis			deflation
		uncorrected	first order	Σ_d	uncorrected	first order	Σ_d	
A	ridge	0.937	0.027	0.043	1.000	0.023	0.043	0.91
A	generalised ridge	0.997	0.030	0.060	1.000	0.017	0.047	0.93
A	Firth	0.067	0.067	0.067	0.053	0.053	0.053	1.03
B, $\kappa = 0.25$	ridge	1.000	0.013	0.057	1.000	0.023	0.077	0.79
B, $\kappa = 0.25$	generalised ridge	1.000	0.023	0.080	1.000	0.027	0.070	0.83
B, $\kappa = 0.25$	Firth	0.987	0.117	0.317	1.000	0.127	0.297	1.15